Experiment 3
Free Fall of Objects

Victoria Torres Ochando and Martina Lodos Falcón
Group A3-6

Laboratory session 23/04/2025

Abstract

The main objective of this experiment is to verify that the acceleration due to gravity g is independent of an object's mass. To achieve this, we determined the local value of g by measuring the fall time of two metallic spheres of different masses from various known heights. Using a free-fall electromagnet system and a high-precision digital timer, the relationship between height and the square of the fall time was analyzed. Linear regressions confirmed the expected proportionality, and the calculated values of g support the theoretical prediction that all objects fall with the same acceleration regardless of their mass. Minor discrepancies with the expected local value are attributed to systematic errors and equipment limitations.

Contents

1 Introduction **1**
1.1 Contextualization 1
1.2 Theorical principles 2

2 Materials and Methods **3**
2.1 Materials 3
2.2 Methods 4

3 Results and discussion **4**

4 Conclusions **7**

5 Appendixes **7**
5.1 Calculation of Uncertainties 7

1 Introduction

1.1 Contextualization

The measurement of gravitational acceleration is one of the most fundamental experiments in Classical Physics and plays a key role in understanding motion and the forces acting on bodies near Earth's surface. The scientist Galileo was the first one to investigate free fall, focusing on determining the local acceleration due to gravity, g, by measuring the fall time of objects released from a known height [1].

One of the central objectives is to verify the mass independence of g, a result predicted by Newton's Second Law in combination with the Law of Universal Gravitation. This allows us to confirm that all objects near the Earth's surface, when subject only to gravity, accelerate uniformly regardless of their mass [2].

1.2 Theorical principles

Free fall refers to the motion of an object when the only force acting on it is gravity. In the absence of air resistance -which is negligible for dense, compact objects like metal spheres over short distances- the object accelerates uniformly towards Earth's surface.

According to Newton's Law of Universal Gravitation, the gravitational force between two masses is directly proportional to the product of their masses and inversely proportional to the square of the distance separating their centers. When applied to an object near the Earth's surface, this force is expressed as:

$$F_g = G\frac{M_T \cdot m}{r^2}(N), \tag{1}$$

where, G is the universal gravitational constant -with a value of $6.47 \cdot 10^{-11} Nm^2kg^{-2}$-, M_T is the Earth's mass, m is the object's mass, and r is the distance between the center of the Earth and the object. Since the height from which objects fall in our laboratory setting is extremely small ($h \simeq 0,738km$) compared to Earth's radius ($R_T = 6370km$), r can be approximated as the Earth's radius R_T. This simplifies the force equation to:

$$F_g = G\frac{M_T \cdot m}{R_T^2}(N). \tag{2}$$

This allows us to define the gravitational field strength (gravitational acceleration) at Earth's surface as:

$$g = G\frac{M_T}{R_T^2}(m/s^2). \tag{3}$$

Substituting g, defined in the previous formula, into equation 2 gives Newton's Second Law,

$$F_g = m \cdot g(N). \tag{4}$$

This relation reveals a key concept: the gravitational force on an object near Earth's surface causes it to accelerate at a rate g, and this acceleration does not depend on the object's mass. Therefore, if we consider only the vertical direction of the object's motion, the mass undergoes uniformly accelerated motion (with g constant).

If we assume the object starts from rest (initial velocity $v_0 = 0$), its velocity and displacement over time are described by the kinematic equations 5 and 6.

$$v = v_0 + g(t - t_0) = g(t - t_0)(m/s), \tag{5}$$

and

$$s = s_0 + v_0(t - t_0) + \frac{g}{2}(t - t_0)^2 = s_0 + \frac{g}{2}(t - t_0)^2(m), \tag{6}$$

where v and v_0 are the final and initial velocity of the object, s and s_0 are the terminal and initial position of the mass and t and t_0 are the final and initial moments.

Considering initial time $t_0 = 0$ s and $h = s - s_o$ m, we obtain the following equation.

$$h = \frac{g}{2}t^2(m)$$

$$g = 2\frac{h}{t^2}(m/s^2) \tag{7}$$

These equations form the basis for determining the value of g experimentally by measuring the time t an object takes to fall a known distance h. Additionally, using two spheres of different mass allows for a direct test of whether mass has any influence on the rate of free fall—an important confirmation of the universality of gravitational acceleration.

In this experiment we will plot $t^2 = t^2(h)$, obtaining g as follows.

$$a = \frac{t^2}{h}, \tag{8}$$

where a is the slope of the graphic, which we can replace for $\frac{t^2}{h}$ (as shown in 8), in equation 7, resulting in:

$$g = \frac{2}{a}. \tag{9}$$

2 Materials and Methods

2.1 Materials

The materials that were used in this practice are the following[2]:

- **Electromagnet free-fall device**. Used to suspend the spheres at the desired height and make them fall, it had a built-in ruler. The latter was accurate to within 0.001 m.
- **Digital timer**. With a resolution of 0.0001s, it was used to determine the time it took for each sphere to fall from a given height. For that, it was connected to a small plate to determine when the sphere finally made contact with the surface.
- **Metal spheres**. Used during practice to determine the value of gravity, each sphere was characterized by a different mass (a weight of 0.0283 kg g for the bigger one 2 and 0.0162 g for the smaller ball 1).
- **Scale**. Used to determine the mass of the spheres, it has a resolution of 0.0001 kg.

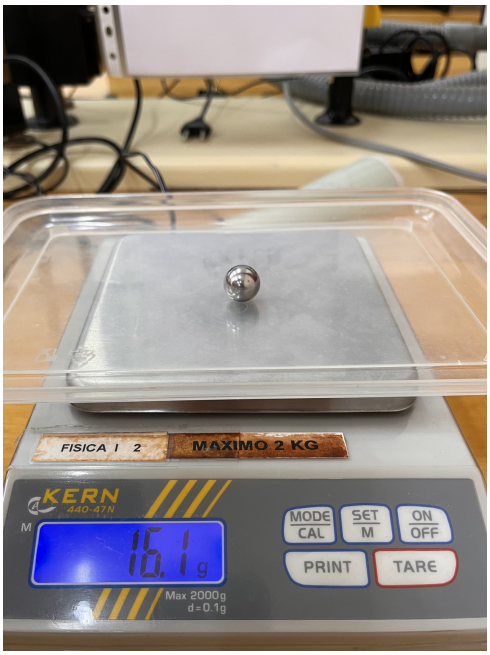


Figure 1: Small ball



Figure 2: Big ball

2.2 Methods

After determining the mass 'm of each sphere using the laboratory balance, we will proceed with the experiment as follows:

1. We will connect all devices and turn on the timer. We will press the 'STOP' and 'RESET' buttons to reset the counter to zero.
2. We will place the adjustable platform at the maximum position (1 m) according to the graduated ruler of the device. Then, we will choose one of the two spheres.
3. We will place the selected sphere on the electromagnet and press 'START' to release it. As it falls, the timer will record the time until the sphere hits the lower plate (fall time), which we will note later.
4. This process will be repeated three times for each sphere at the same height.
5. From that we will decrease that height in increments of 0.05 m (from 1 m to 0.4 m) and repeat the measurements three times as said.
6. Finally, with the collected data, we will proceed to calculate the results, including the value of the local gravitational acceleration 'g' using equation 9 and, to find 'a', equation 8, among others.

3 Results and discussion

The data obtained from throwing the mass of $0,028300 \pm 0,000029$ kg is organised in 1, whereas the information from this experiment for a mass of $0,016200 \pm 0,000029$ kg is shown in 2 .

$h(m)$	$\bar{t}(s)$	$u_C(\bar{t})(s)$	$\bar{t}^2(s^2)$	$u_C(\bar{t}^2)(s^2)$
$1,00000 \pm 0,00029$	0,450700	0,000065	0,203130	0,000058
$0,95000 \pm 0,00029$	0,43893	0,00014	0,19266	0,00012
$0,90000 \pm 0,00029$	0,427167	0,000073	0,182471	0,000062
$0,85000 \pm 0,00029$	0,414967	0,000044	0,172197	0,000037
$0,80000 \pm 0,00029$	0,40200	0,00016	0,16160	0,00012
$0,75000 \pm 0,00029$	0,38890	0,00016	0,15124	0,00012
$0,70000 \pm 0,00029$	0,3689	0,0067	0,1361	0,0049
$0,65000 \pm 0,00029$	0,36153	0,00012	0,130706	0,000089
$0,60000 \pm 0,00029$	0,347067	0,000044	0,120455	0,000031
$0,55000 \pm 0,00029$	0,33220	0,00021	0,11036	0,00014
$0,50000 \pm 0,00029$	0,31593	0,00012	0,099814	0,000078
$0,45000 \pm 0,00029$	0,29893	0,00014	0,089361	0,000082
$0,40000 \pm 0,00029$	0,28147	0,00015	0,079223	0,000083

Table 1: $\bar{t}$ is the average time t_i (with $i = 1, 2, 3$) for each height h, and $u_C(\bar{t})$ is its combined uncertainty; and $\bar{t}^2$ and $u_C(\bar{t}^2)(s)$ refers to the squared average time and its uncertainty for the heaviest ball.

$h(m)$	$\bar{t}(s)$	$u_C(\bar{t})(s)$	$\bar{t}^2(s^2)$	$u_C(\bar{t}^2)(s^2)$
$1,00000 \pm 0,00029$	0,44983	0,00026	0,20235	0,00011
$0,95000 \pm 0,00029$	0,43917	0,00012	0,192867	0,000048
$0,90000 \pm 0,00029$	0,42700	0,00027	0,182329	0,000097
$0,85000 \pm 0,00029$	0,414333	0,000044	0,171672	0,000015
$0,80000 \pm 0,00029$	0,40240	0,00025	0,161926	0,000082
$0,75000 \pm 0,00029$	0,38897	0,00020	0,151295	0,000062
$0,70000 \pm 0,00029$	0,37527	0,00035	0,140825	0,000098
$0,65000 \pm 0,00029$	0,36210	0,00016	0,131116	0,000041
$0,60000 \pm 0,00029$	0,34703	0,00024	0,120432	0,000057
$0,55000 \pm 0,00029$	0,331267	0,000093	0,109738	0,000020
$0,50000 \pm 0,00029$	0,31553	0,00029	0,099561	0,000057
$0,45000 \pm 0,00029$	0,29960	0,00042	0,089760	0,000075
$0,40000 \pm 0,00029$	0,280733	0,000044	0,0788112	0,0000070

Table 2: $\bar{t}$ is the average time t_i (with $i = 1, 2, 3$) for each height h, and $u_C(\bar{t})$ is its combined uncertainty; and $\bar{t}^2$ and $u_C(\bar{t}^2)(s)$ refers to the squared average time and its uncertainty for the lightest ball.

With this information we are able to plot graphic 3 and 4 for $t^2(h) = t^2$ using the least squares method.

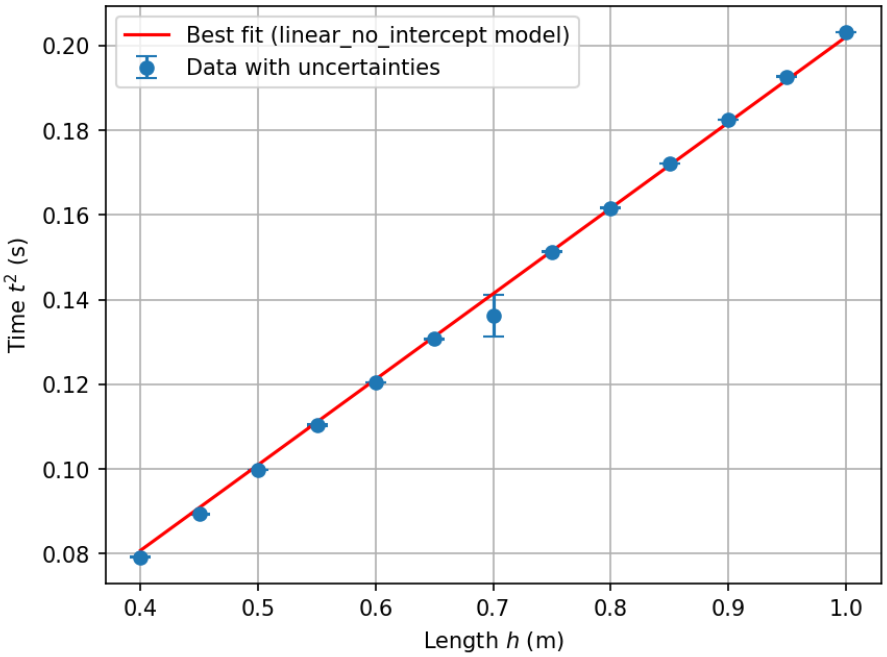


Figure 3: Free fall of a mass of $0,028300 \pm 0,000029kg$.

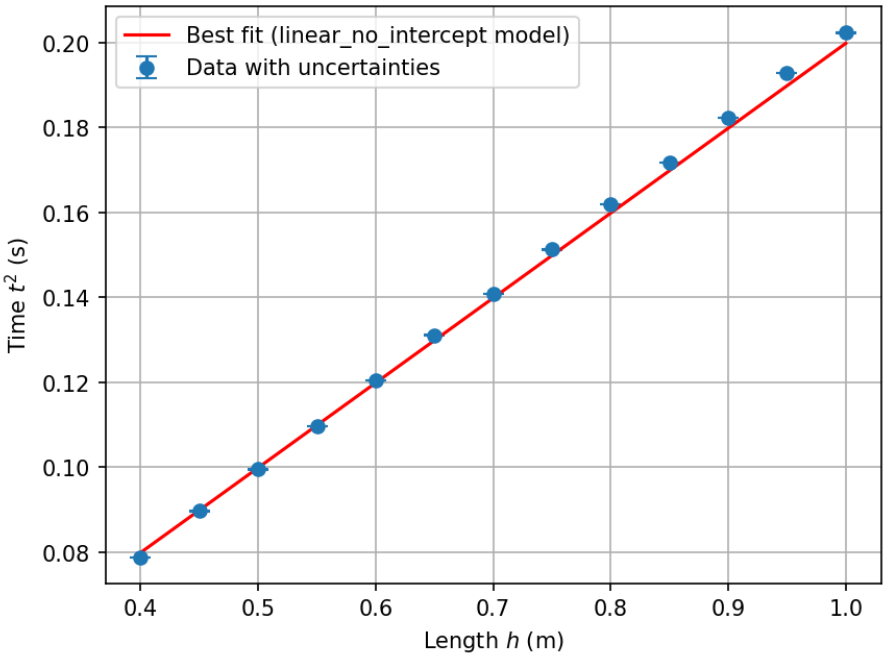


Figure 4: Free fall of a mass of $0,016200 \pm 0,000029kg$.

Clearly, the time squared t^2 has a linear relationship with the height h, which confirms the theoretical prediction of kinematic equations for constant acceleration g.

The slope is defined as explained in equation 8, and has a value of 0.202003 ± 0.000024 for the first ball, and of 0.199837 ± 0.000011 for the second one. Using relation 9 we obtain the following results.

$m(kg)$	$a(s^2/m)$	$u_C(a)(s^2/m)$	$g(m/s^2)$	$u_C(g)(m/s^2)$	r^2
$0,028300 \pm 0,000029$	0,202003	0,000024	9,90099	0,00123	0,99942
$0,016200 \pm 0,000029$	0,199837	0,000011	10,0100	0,0013	0,99997

Table 3: m is the mass of each ball; a is the slope of the graphics and $u_C(a)$ its uncertainty; g is the value of the gravitational acceleration; and r^2 is Pearson's correlation coefficient.

As can be seen in 3, Pearson's correlation coefficient is nearly 1, which indicates that there is a high level of confidence in our results. Furthermore, the values of g calculated from experimenting with two different masses are quite similar, meaning that, as exposed in the Theorical principles section, the acceleration of an object in free fall does not depend on its mass.

However, the values of g obtain are somewhat larger than expected, since the gravitational acceleration in Granada is $9,796933 \pm 0,000052$, according to [3].

4 Conclusions

This experiment confirmed that the gravitational acceleration g of an object in free fall is independent of its mass m, as shown by consistent g values for two different spheres, $9.9010 \pm 0.0012 m/s^2$ and $10.0100 \pm 0.0013 m/s^2$. Despite slight deviations from Granada's theoretical value ($9.7969 m/s^2$) the results validate the theory with high precision. Systematic errors likely caused the offset, but the core principle, uniform acceleration in free fall, was successfully demonstrated. In conclusion, the experiment effectively bridges theory and, although improved calibration could enhance accuracy.

5 Appendixes

5.1 Calculation of Uncertainties

As we know, type A uncertainty, $u_A(x)$, which is associated to human error, can be calculated with the following equation:

$$u_A(x) = \frac{\sigma(x)}{\sqrt{N}}, \tag{10}$$

where $\sigma(x)$ represents the standard deviation and N the number of observations.

On the other hand we get type B uncertainty, $u_B(x)$, related to the measuring instrument error from:

$$u_B(x) = \frac{\delta_x}{\sqrt{12}}, \tag{11}$$

where δ_x is the resolution of the measuring instrument (δ_h=0,001 m)

From them, $u_C(x)$ is expressed as:

$$u_C(x) = \sqrt{u_A^2(x) + u_B^2(x)}. \tag{12}$$

Additionally, we can calculate the expanded uncertainty is calculated to overestimate the error. To do so, we multiplied $u_C(x)$ by a coverage factor kp that is selected for convenience:

$$U_C(x) = u_C(x) k_p. \tag{13}$$

This factor is derived from the distribution of the t-student, using $n-1$ degrees of freedom and a fixed confidence level. In this case, consistently 95%:

$$kp = 2.178813.$$

So, for all heights, we get:

$$u_C(h)(m) = \sqrt{u_A^2(h) + u_B^2(h)}(m). \tag{14}$$

And, for all calculus of $\bar{t}$:

$$u_C(t)(s) = \sqrt{u_A^2(t) + u_B^2(t)}(s). \tag{15}$$

Regarding the error of the calculated t^2, since it is an indirect measurement obtained from t, we express:

$$u_C(t^2)(s^2) = \sqrt{\left(\frac{\partial t^2}{\partial t}\right)^2 (u_C(t))^2} = \sqrt{(2 \cdot t)^2((u_C(t))^2} = 2t \cdot u_C(t)(s^2). \tag{16}$$

To continue, the aceleration of gravity, g, obtained indirectly from equation 9, obtains an uncertainty $u_C(g)$:

$$u_C(g)(m/s^2) = \sqrt{\left(\frac{\partial g}{\partial a}\right)^2 (u_C(a))^2} = \sqrt{\left(\frac{-2}{a^2}\right)^2 (u_C(a))^2} = \frac{2}{a^2} \cdot u_C(a)(m/s^2). \tag{17}$$

References

[1] *Motion of Free Falling Object*. July 21, 2022. URL: https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/motion-of-free-falling-object/#:~:text=An%20object%20that%20falls%20through,Newton's%20second%20law%20of%20motion..

[2] I. Adroher; D. Bianco; A. Cazoría; J.R. Morilla; F. Navas; R. Rica; S. Koldan; G. Titos; A. Valenzuela. *Laboratory Manual for Basic Experimental Techniques*. Department of Applied Physics, University of Granada, 2025.

[3] Enrique Rodríguez Pujol. "Medidas gravimétricas en Madrid y en España". In: *Instituto Geográfico Nacional* (2017).